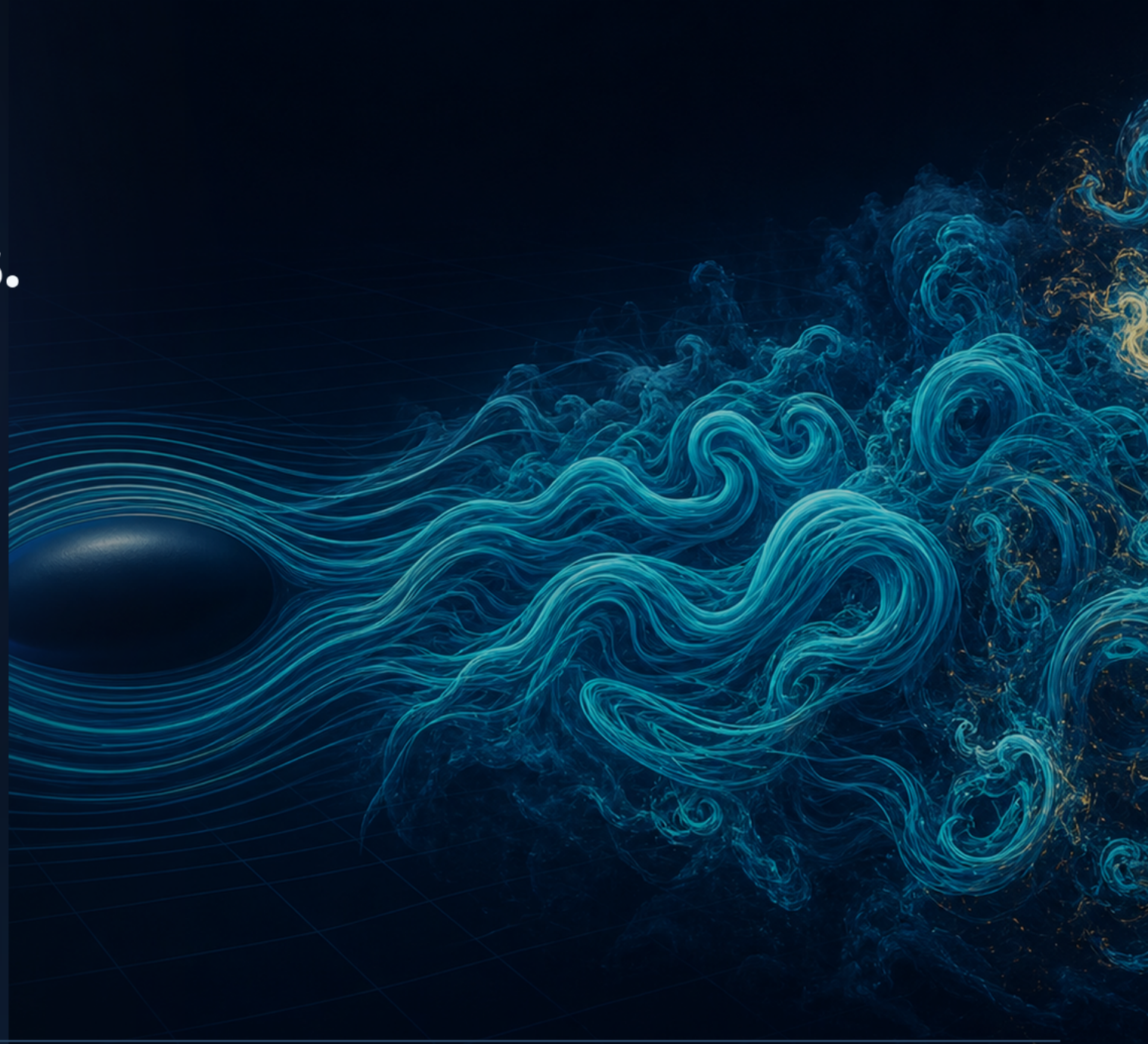
PUBLIC RESEARCH GUIDE

HUMAN-DIRECTED • AI-EXECUTED

# Fast falsification. Shorter research loops.

Under human direction, Codex turns proof gaps into C++ tests, attacks weak lemma routes, reads the failures and carries the surviving structure into the next candidate.



PRIMARY ADVANTAGE

## Bad directions fail quickly.

- C++ searches stress proposed lemmas on finite models.
- Lean checks exact counterexamples to false bounds.
- AI uses each failure to refine the next proof obligation.

CURRENT ACTIVE LAB

## 3D Navier–Stokes existence and smoothness

Finite formal results and research code — not a Clay solution.

# Seven problems, one active laboratory

The repository is organized by Millennium Prize Problem. Navier–Stokes is active; the other unsolved tracks are explicit future scope.

| PROBLEM | AREA | CLAY STATUS | REPOSITORY |
|---|---|---|---|
| **Birch–Swinnerton-Dyer** | Number theory | NOT STARTED | Planned |
| **Hodge conjecture** | Algebraic geometry | NOT STARTED | Planned |
| **Navier–Stokes** | PDE / fluid dynamics | ACTIVE | C++ and Lean audit |
| **P vs NP** | Complexity theory | NOT STARTED | Planned |
| **Poincaré conjecture** | Topology | SOLVED EXTERNALLY | No lab needed |
| **Riemann hypothesis** | Number theory | NOT STARTED | Planned |
| **Yang–Mills mass gap** | Mathematical physics | NOT STARTED | Planned |

Official problem definitions and current solved/unsolved classification: claymath.org/millennium-problems

RUN IN CAMPAIGN

# AI executes the research loop

Under human direction, Codex formulates candidates, writes and runs the C++, interprets failures and designs the next lemma. Every claim returns to exact tests and expert proof review.

INPUT 01

### Proof obligation

A precise inequality, scaling law or closure target from the proof plan.

IMPLEMENTED

INPUT 02

### Empirical artifacts

TSV states, JSON certificates, gradient traces and failed-lemma records.

IMPLEMENTED

ENGINE 03

### Adversarial search

Exact objectives, JVP/VJP checks, L-BFGS and multistart stress the statement.

IMPLEMENTED

SYNTHESIS 04

### AI pattern analysis

Read recurring geometry, scaling and failure modes across machine artifacts.

RUN IN CAMPAIGN

OUTPUT 05

### New candidate lemma

Formulate a narrower statement with explicit assumptions, constants and tests.

AI-REFINED

THE HUMAN-DIRECTED, AI-EXECUTED RESEARCH LOOP

1

**FORMULATE**

Write a lemma with a measurable finite analogue.

2

**ATTACK**

Search for the worst state, not a reassuring example.

3

**FALSIFY OR SURVIVE**

Store failure evidence or a bounded-looking branch.

4

**EXPLAIN**

Extract the geometry and normalization that caused the result.

5

**PROVE**

Return to cutoff-independent mathematics and peer review.

Lean now proves finite counterexamples; a continuous PDE theorem still needs separate analysis.

FORMAL AUDIT

# Lean verifies counterexamples to unrestricted PNT bounds

A six-mode Fourier family disproves the proposed uniform PNT-2/7/8/12/14 bounds. The result is exact but finite-dimensional.

LEAN-CHECKED COUNTEREXAMPLE

## The unrestricted PNT bounds fail

Lean proves the witness is admissible and computes its full cutoff-one Fourier output. Let $a > 0$ be its amplitude.

EXACT SCALING

$$PNT\text{-}2 = 1 / (876024\, a)$$

PNT-7/8/12/14 grow as $1 / a^2$

WHAT THE RESULT RULES OUT

As a approaches zero, no finite uniform bound survives. Fixed-energy searches ask a different question.

ALSO CHECKED IN LEAN

## Finite identities and inequalities

The package verifies the Cauchy identities and a general finite Gram-Schur estimate. It audits the theorems and tests deliberate proof failures.

| EVIDENCE | RESULT | MEANING |
|---|---|---|
| **Fourier witness** | 6 modes | Full cutoff-one cube |
| **Proof library** | 75 theorems | 259 declarations audited |
| **Negative tests** | 4 rejected | False or unfinished proofs |

VERIFY `./navier-stokes/lean/verify.sh`



No fixed-energy supremum or continuous-PDE regularity is proved

# Research status after the Lean audit

Formal finite counterexamples retire the unrestricted PNT bounds. The continuous Navier–Stokes regularity argument remains open.

**01 PROVED**
### Lean PNT audit
Unrestricted PNT-2/7/8/12/14 bounds are false; finite Schur is proved.

→

**02 OPEN**
### PNT reformulation
Re-derive the admissible set or a homogeneous normalization.

→

**03 OPEN**
### Local closure
Validate the sign candidate and close gap-zero and transition terms.

→

**04 OPEN**
### L4 regularity
Convert the local decomposition into the required trajectory estimate.

→

**05 LATER**
### L5 to L6
Complete the remaining regularity and limit-passage steps.

## NEXT ENGINE MILESTONES

| WORK ITEM | PURPOSE | EXIT CONDITION |
|---|---|---|
| **Exact sign check** | Test the local derivative candidate | Certified sign or rejected claim |
| **PNT reformulation** | Separate fixed-energy from unrestricted claims | Valid lemma or failed route |
| **Analytic closure** | Control the local and transition blocks | Cutoff-independent estimate |

Lean proofs: navier-stokes/lean/README.md • Run: ./navier-stokes/lean/verify.sh

## NON-CLAIM

## The Clay problem is not solved.

The Lean result concerns a finite Fourier model and proposed PNT bounds. A solution still needs the continuous local estimates and the remaining L4–L6 argument.

**RESEARCH SOFTWARE**

**Evidence generator • falsifier • proof navigator**